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Testing of Automotive Systems (Part I)

Module 6 –EE System Testbenches

Martin Lamprecht, Magna Steyr

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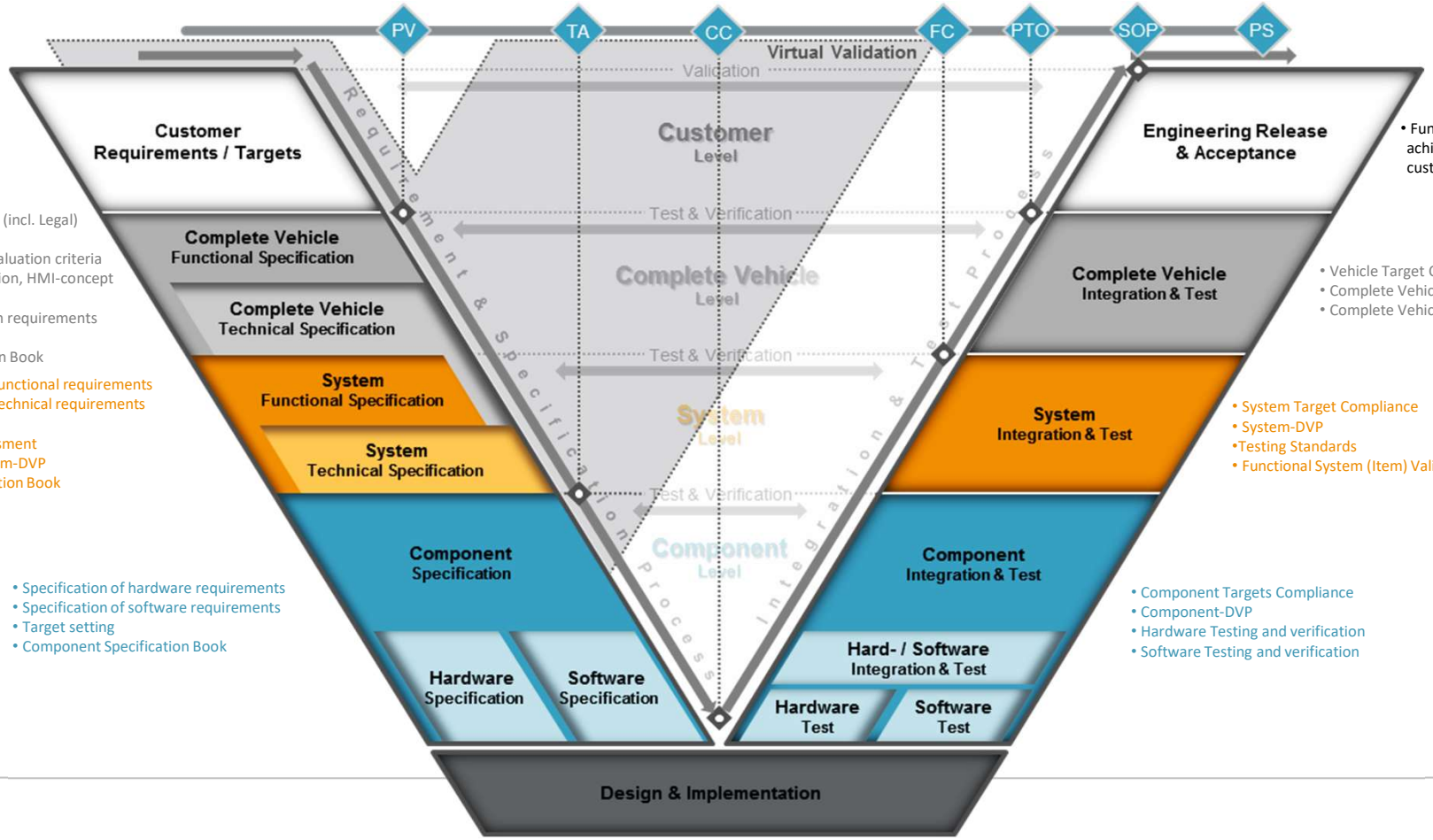
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- Customer Market Profile
- Sales Markets
- Configuration List
- Feature und Function List
- Benchmarks
- Vehicle Profile Description

- Definition of CV requirements (incl. Legal)
- Target setting
- Derivation of measureable evaluation criteria
- Product & Functional description, HMI-concept
- Creation of CV-DVP
- Definition of Items and system requirements
- Item List
- Complete Vehicle Specification Book

- Specification of functional requirements
- Specification of technical requirements
- Target Setting
- Functional assessment
- Creation of System-DVP
- System Specification Book
- Drawings
- Serial BOM

- Specification of hardware requirements
- Specification of software requirements
- Target setting
- Component Specification Book



- Functional target achievement confirmed on customer-ready vehicles

- Vehicle Target Compliance
- Complete Vehicle-Test Program
- Complete Vehicle Function Validation

- System Target Compliance
- System-DVP
- Testing Standards
- Functional System (Item) Validation

- Component Targets Compliance
- Component-DVP
- Hardware Testing and verification
- Software Testing and verification

ECU Testing

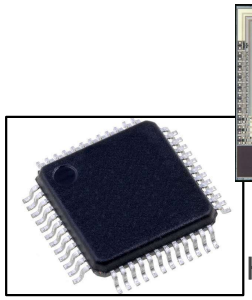
Hardware

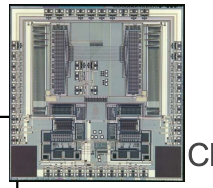
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HW Standards : Different level definitions in development with focus hardware

Tier 2



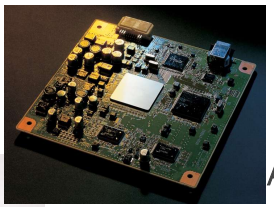


Chip level

Hardware part level

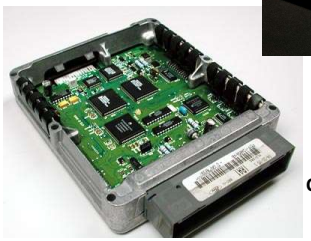
Main qualification Spec: AEC-Q

Tier 1



Assembly level

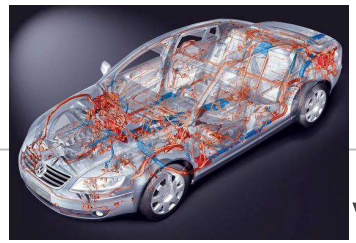
Main qualification Spec: Set of IPC



Component level

Main qualification Spec: ISO-16750 or OEM equivalent

OEM



Vehicle/system level

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Definition of A/B/C/D/E - Samples

A-Sample (Engineering Sample)

Main purpose is to show basic communication and some base functionality is already implemented. It could be either a development board or a design intent housing already

B-Sample (DV-Sample)

All functionality shall be implemented already. All required environmental and EMC tests shall be done.

C-Sample (Off-tool Sample)

100% Off-tool parts – all functionality implemented as well as all **Design Verification (DV) t**asks shall be finished.

D-Sample (PV-Sample)

100% Off-tool and Off-process parts - used to deliver the parts for **Product Validation (PV)** and **Production Part Approval Process (PPAP)**

E-Samples (Series Parts)

Serial production parts with continuous conformity testing (CC)

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Sample Description

Sample-Category	Use / Purpose	Quality	Software	Manufacturing	Conditions for approval	Testing
A	First build up of the concept in hardware for the representation of the customer related functions. Object for FMEA.	Customer related functions confirmed. Deviations to specification / functional restrictions are stated in the test report.	Restrictions of functions according specification. Deviations are stated in release notes which are added to the test report.	Hand- work, special manufacturing, modified processes. Partly alternative materials. Rapid prototyping parts. Tacking according internat. standards.	100% testing of characteristics defined in the prototype control plan	Start of testing at MAGNA in special environment with qualified staff. Hardware is not shipped to the customer.
B	Final hardware. Used for DV testing in an extent that the release of the series tools manufacturing can take place.	Functions confirmed. DV test finalized with positive results. Reliability and lifetime confirmed with defined confidence. Surface not finalized in tools. Deviations to specification / functional restrictions are stated in the test report.	All functions confirmed, which are necessary for the DV testing of the hardware and the operation of the vehicle. All important safety functions confirmed primarily. Deviations are stated in release notes which are added to the test report.	Series related PT building, auxiliary tools (e.g. aluminum or unhardened steel) and auxiliary devices / fixtures. Target: 100% final material.	100% testing of characteristics defined in the prototype control plan. Prototype process flow chart and control plan supplied	Design verification Testing in vehicles. Defined restrictions for operating
C	Confirmation of final design with parts off tools. Samples for building up the assembly line. Internal release of production process at supplier	Same as B. All functions of the hardware according specification confirmed. Can be flashed in serial production process	Same as B, but all specifications confirmed. All safety functions confirmed.	100% off tool (not hardened) 100% final material Start-up of series process in series production site	100% testing of characteristics defined in the series control plan. Series process flow chart and control plan supplied	Testing with parts off serial tools. Testing in vehicles. Defined restrictions for operating
D	PV testing PPAP parts Run at rate Release / approval	Same as C Reliability and lifetime confirmed Short-term process capability confirmed. Parts marked like parts from serial production.	Same as D. Documentation finalized.	100% final process at 100% cycle time 100% final tools 100% final material 100% series staff 100% series packing MSA done; IMDS done Parts from / after run@rate	Documentation for process and product approval completed	Product validation Assembly in pre-series vehicles
Series P	Serial production Continuous Conformity Testing	Same as D confirmation of long term process capability	Same as D	Same as D	Same as D confirmation of long term process capability	Assembly in serial vehicles for the customer

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Test Item	Test against	Target Summary	Main Characteristics	Test Bench
Hardware Unit and Integration Testing (HWIT)	Hardware Detailed Design	Verify HW design decisions, functionality	White-box, low level, Hardware	Basic measurement tools, normal working place
	Hardware Architecture Spec	Interfaces between HW Units / Components	White-box, Hardware Component level	Basic measurement tools, normal working place
Hardware Tests (HWT)	HW Requirement Spec	Verify requirements	Mainly black-box	Normal working place HiL Special Lab (climatic, mechanical, chemical chambers
Qualification of HW Comp.	Integrated HW components	Suitability for usage	Testing (functional, climatic…) Analysis (simulation, extrapolation…)	Special Lab (climatic, mechanical, chemical chambers Computer workplace

HW Development: Hardware Unit and Integration Testing (HWIT)

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Verification of:

Hardware Architecture and Detailed Design Specification

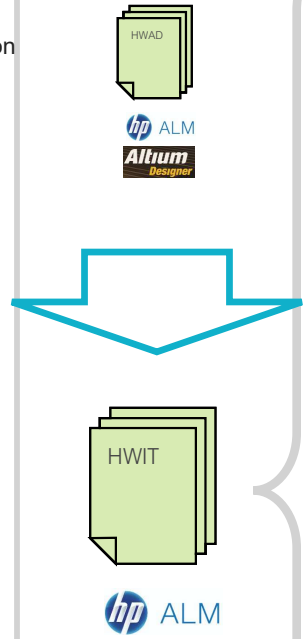
Hardware Construction

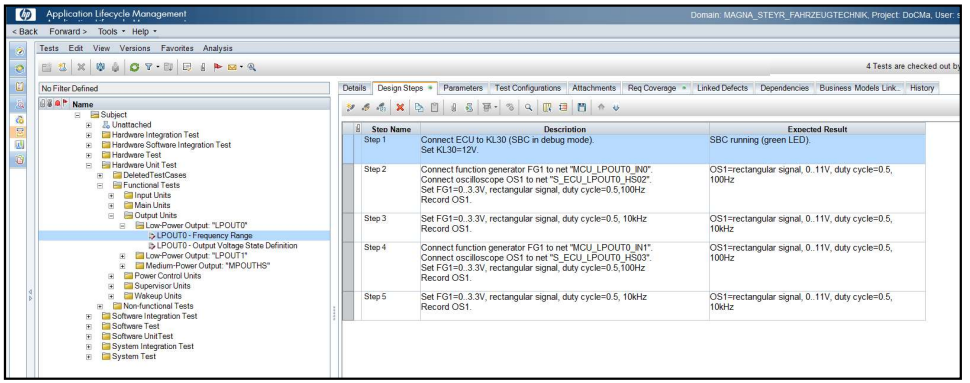
Tasks:

- White Box Testing
- Datasheet Limit/Worst Case Conditions
- Design Decision Limits/Worst Case Conditions
- Electrical Testing
- Error Guessing
- Fault Injection Testing
- Unit Tests
  - HW Functional Testing
  - HW Expanded Function Testing
- Integration Tests
  - Internal and external Interfaces Functional Testing
  - Expanded Interfaces Functional Testing

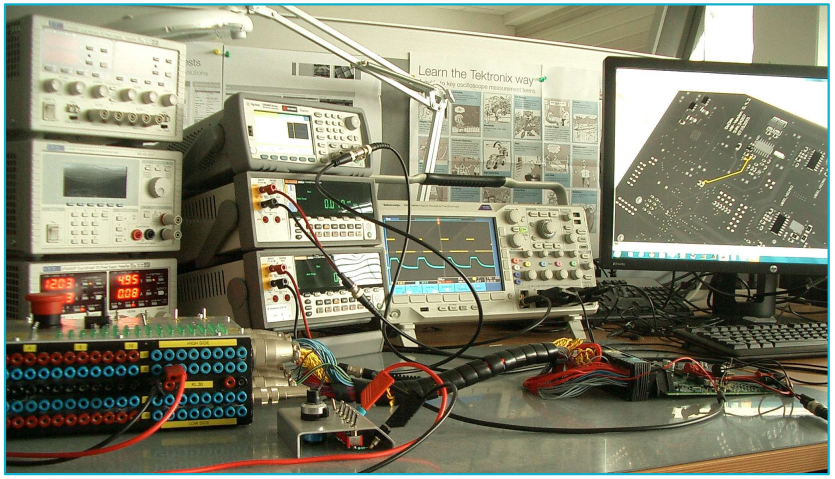
Examples:

- HP ALM Test Specification and Test Steps
- HP ALM Test Results





HP ALM: Hardware Unit Test Specification for Low-Power Output “LPOUT0”. Extract with 1 test case and its steps.



HW Unit Test: Measurement setup for frequency range upper limits of medium power output driver.

HW Development: Hardware Testing (HWT)

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Verification of:

Hardware Requirement Specification (result of Hardware Requirement Analysis)

Tasks:

- Test Plan and Test Specification
- Black Box Testing
- Electrical Testing according to OEM/internal standards
- Environmental Testing according to OEM/internal standards

Examples:

- Hardware Qualification Test Plan Overview XGW
- Hardware PV/DV Test Report XGW
- DIAdem Relay Test Issue Report XGW

HWRA

ALM

HWT

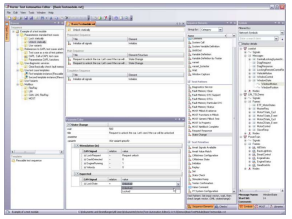
vector
VT System

NATIONAL INSTRUMENTS
DIAdem

ALM
Under construction



HiL Testing: Test Bench for electrical Tests



HiL Testing: Test Automation Editor



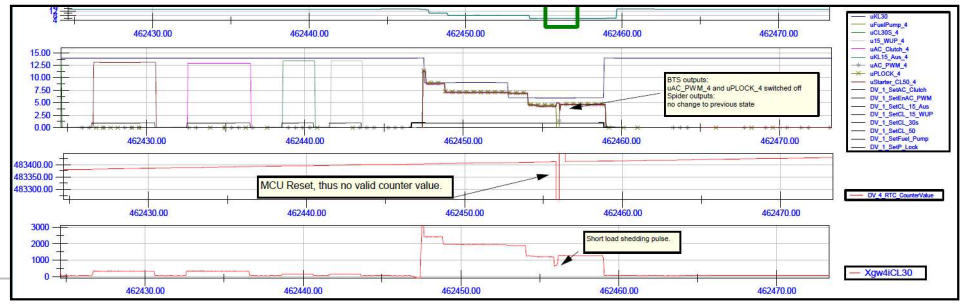
PV/DV Test s: Climate Chamber



PV/DV Tests: Simulation Boards



PV/DV Tests: Devices under Test



PV/DV Test Results Verification: NI DIAdem Report Extract

HW Development: Hardware Testing (HWT) – Test plan

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Verification of:

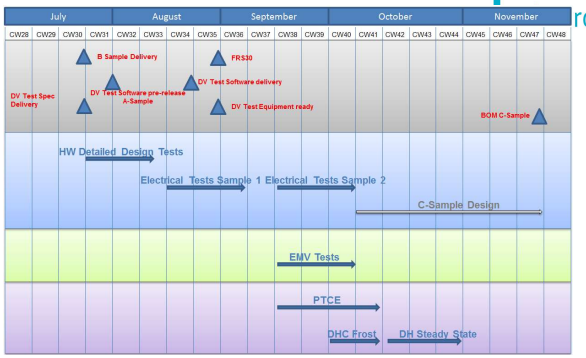
Hardware Requirement Specification (result of Hardware Requirement Analysis)

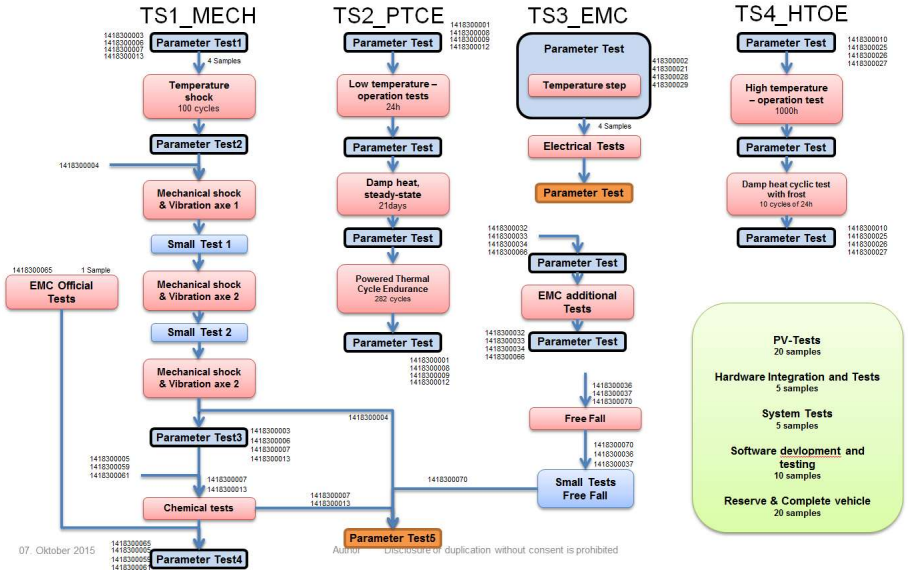
Tasks:

- Specify list of tests to be executed on each sample
- Arrange test samples
- Schedule tests according to development schedule
- Plan and supervise testing budget with PM
- Coordinate testing with external test labs

Examples:

- Hardware Qualification Test Plan Overview XGW





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Content - Backup - References

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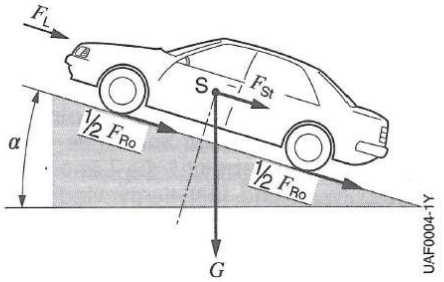
ECU Testing

Hardware in the loop - HiL

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EE Testing – Concepts

- Hardware in the Loop and Simulation-based testing
  - testing of electrical functions in an early project phase
  - reproducible testing, automated testing
  - testing is possible long before vehicle prototype is available
  - fault injection made easy



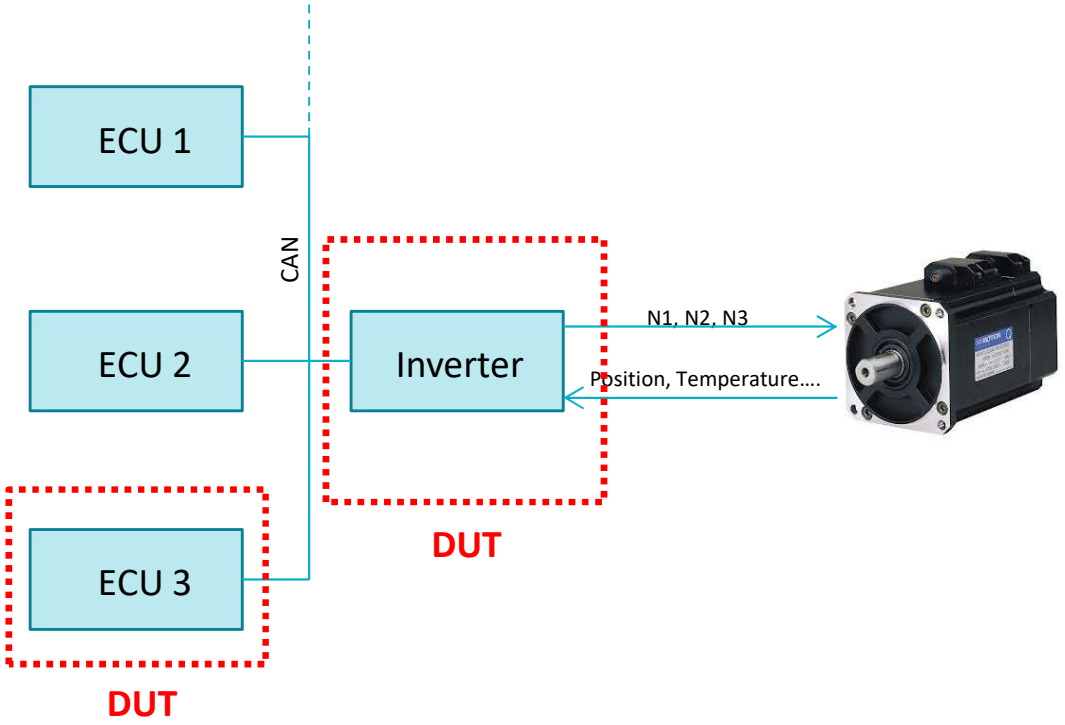
- HiL benefits:
  - "Virtual Test Drives", ECUs connected to a "mathematical model"
  - environmental conditions change on a click (temperature/slope/friction)

- Simulation benefits:
  - no complex mathematical models needed
  - easy to extend



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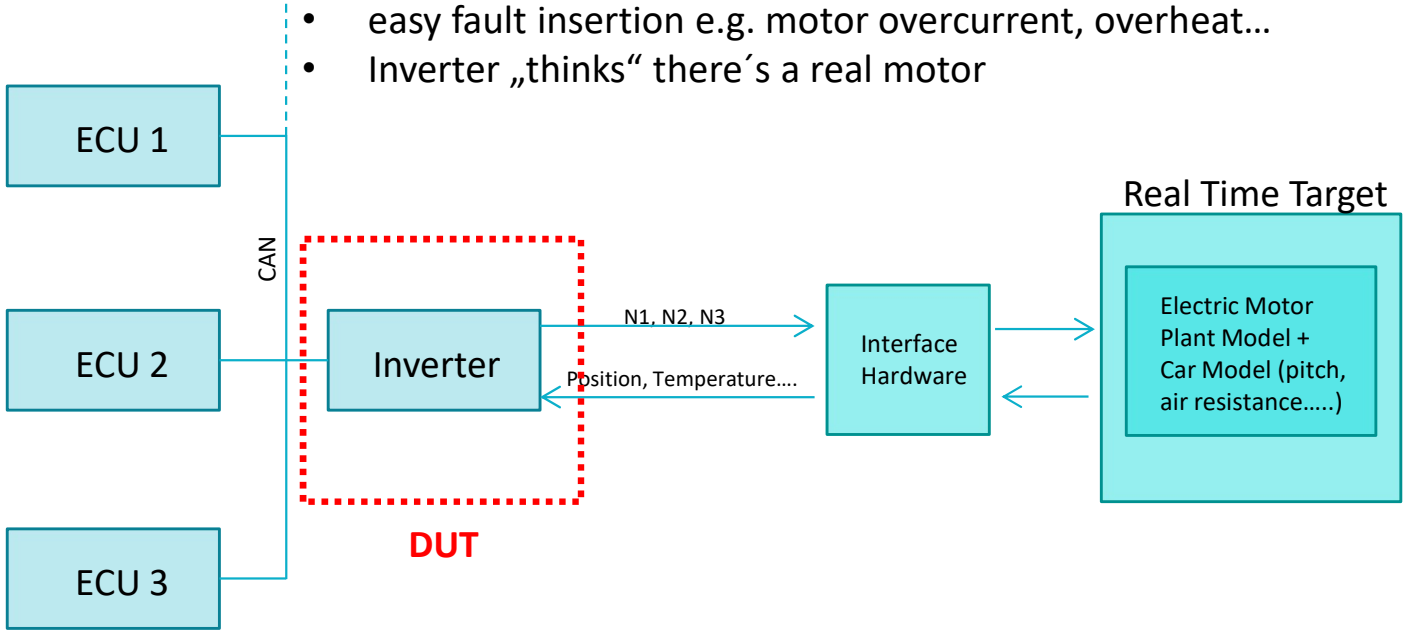
System example - real configuration



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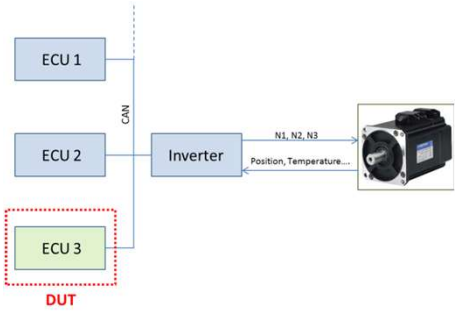
Hardware In the Loop - configuration

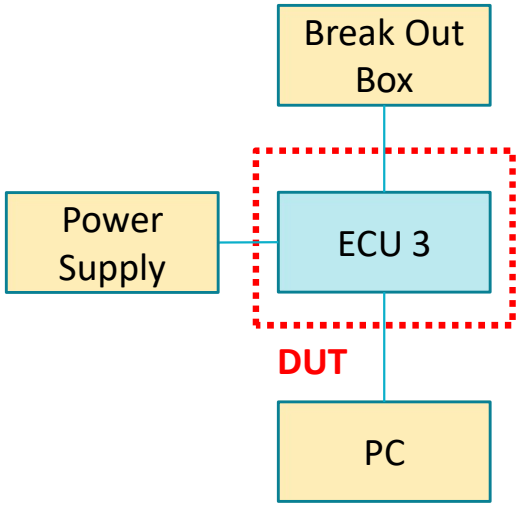
- **Complete electrical system of the car available**
- Mathematical Model of simulated hardware part
- Test bench testing without real electric motor (no moving parts)
- easy fault insertion e.g. motor overcurrent, overheat…
- Inverter „thinks“ there´s a real motor



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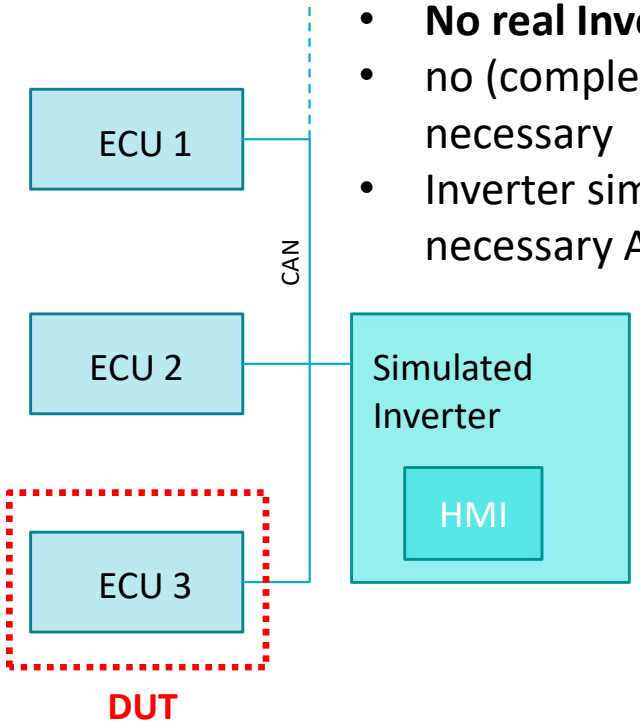
HiL configuration on ECU level





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Hardware In the Loop – simulation configuration



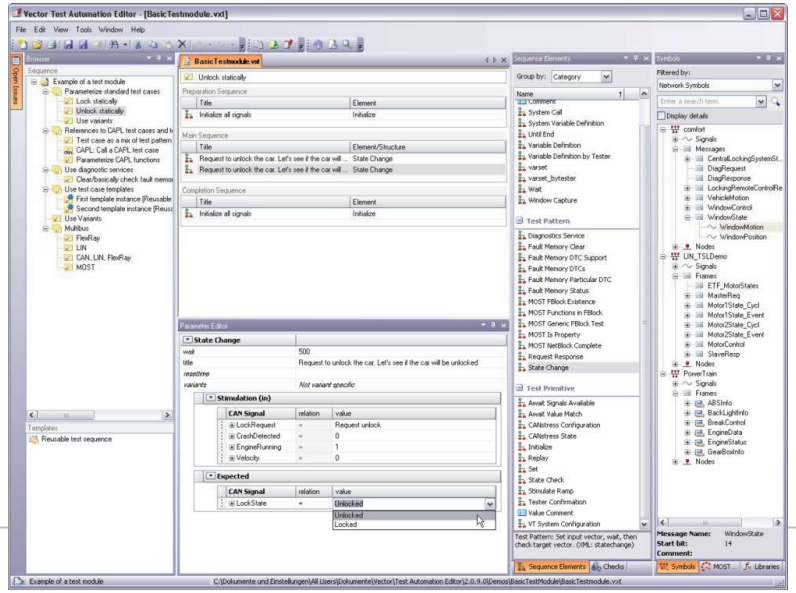
- **No real Inverter**
- no (complex) mathematical models necessary
- Inverter simulated on bus level, only necessary A/D signals generated

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Hardware In the Loop - Tools used

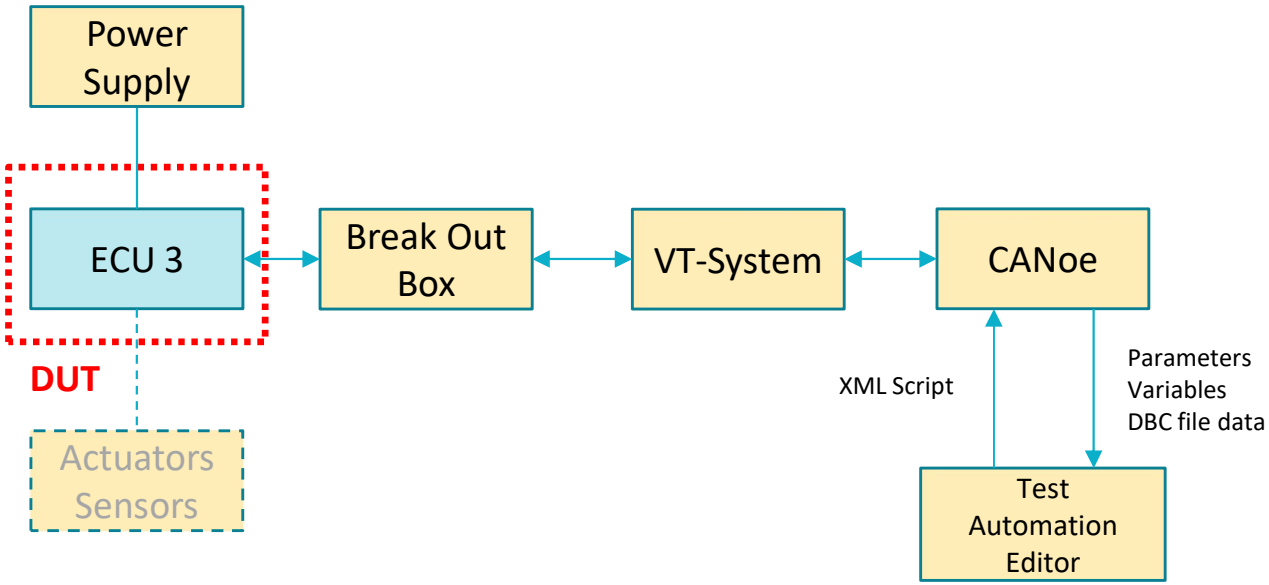
- **Test Automation Editor** - development environment for creating automated ECU tests
- **CANoe** - software tool for development, test, and analysis of entire ECU networks and individual ECUs (LINK)
- **VT System** - Modular Test Hardware comprising bus interfaces, measurement modules, stimulation modules, relay modules…





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HiL configuration on ECU level



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Input documents and work products

» Input Documents

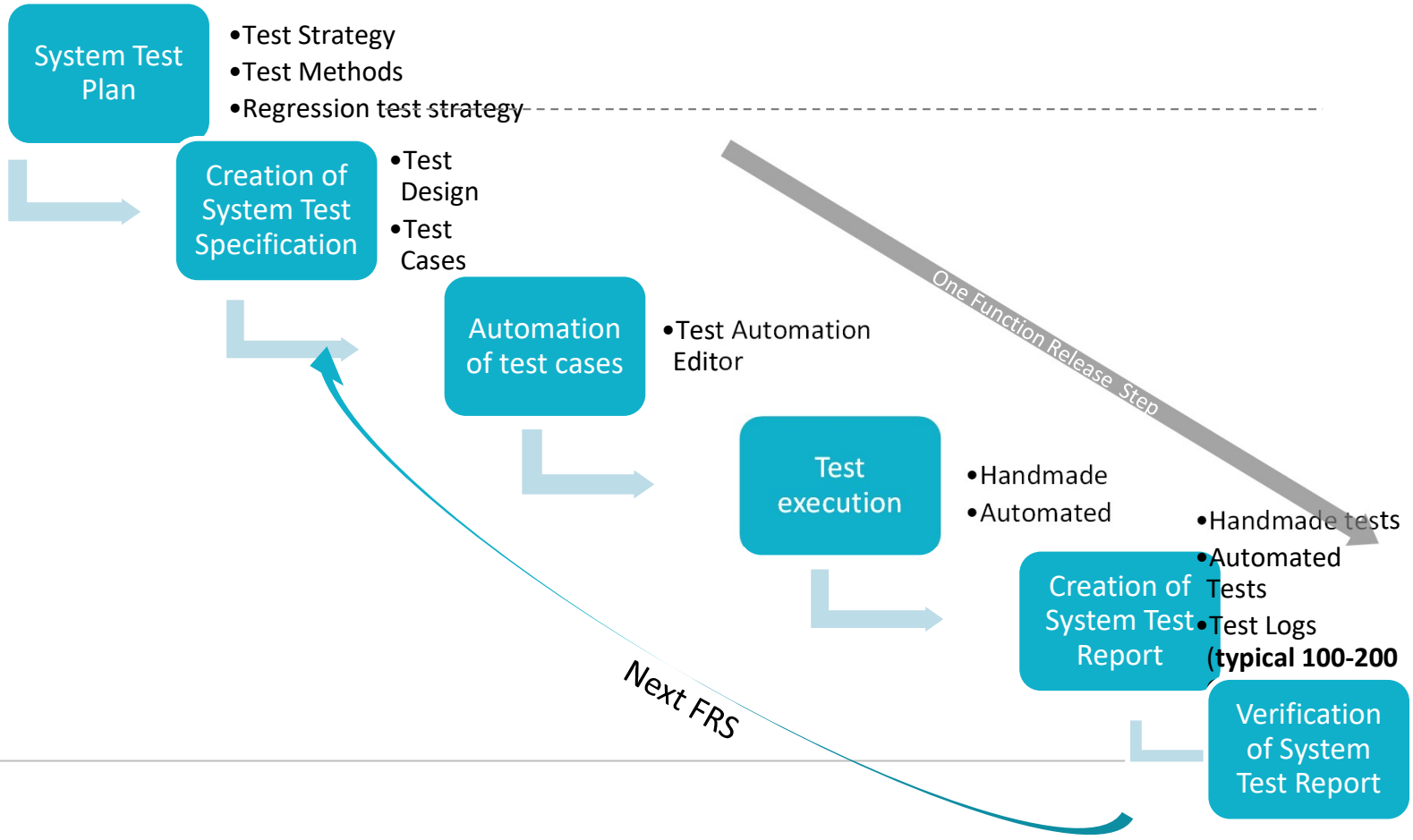
- **System Requirement Specification**- defined and refined customer requirements/functions on system level
- **Standards (IEEE, ISO ….)** – standardized communication protocols or standardized diagnostic communication in automotive

» Work products

- **System Test Plan**- defined Test strategy including test methods, test design technics and regression test strategy
- **System Test Specification**- including test case specification, test design specification and test environment specification
- **System Test Report-** Results of all accomplished tests (automated and hand made)
- **Traceability Record-** All tests have to be traceable to the input document and the next work product

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Hardware In the Loop - example



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Test Plan

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4 System Test

4.1 Test Goal

The purpose of the Systems testing process is to ensure that the implementation of each system requirement is tested for compliance and that the system is ready for delivery.

4.2 Test Strategy

4.2.1 Planning and Organization

The test schedule for A-, B-, C-, D-Sample releases can be found in the XGW work product status document APStatus [37].

The project roles for performing the tests can be found in the project role matrix document [38].

Initially, the system requirements specification must be analyzed. The requirements must be prioritized for the impact of these requirements on the operational state of the complete J11 system. Additionally, the requirements must be categorized in safety and non safety relevant requirements and in functional parts of the system.

For each system requirements a sufficient number of test cases will be defined and documented in the system test specification. The test cases must be developed to demonstrate compliance with the system
Requirements, including the interfaces between system elements. The order of the sequence of the defined test cases shall be derived from priorities and categorization the system requirements in order to find errors earlier in the system development life cycle.

The consistency and bilateral traceability between system requirements and the system test specification including test cases are established by the tool REqTracer. Consistency check is done by review.

The test cases in the test specification includes input test vectors as well as expected results, i.e. output vectors.

4.2.2 Testing Methods

The System test will be done in Black-Box- Testing Method.

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Test Specification

» Document examples

No.	Test case name	SYSTS0094
	Test case outline	Function test Clamp_50 emergency start
Prio.	Expected test result outline	Clamp_50 activate with emergency start conditions
	Time involved (time effort)	

Traceability

SYSR0159

Detailed description

Clamp_50_IK (active high HW input)
Clamp_15_1 (active high HW input)
RPM_ENG (PT-CAN, Torque_3)
ST_ENG_RUN (PT-CAN, ENGINE_1)

	Test steps:	Test step result
1.	**Clamp_15_1** (Analog in)< 3 V and **Clamp_50_IK** = 0	Clamp_50=0
2.	**Clamp_15_1** (Analog in)> 7,5 V and **Clamp_50_IK** = 0	Clamp_50 = 0
3.	**Clamp_15_1** (Analog in)> 7,5 V and **Clamp_50_IK** = 1 (min. 300ms) and vehicle speed of is > 10 km/h (**V_VEH** > 10) and Engine is not running (**ST_ENG_RUN**~= ST_ENGINE_RUN)	Clamp_50 = 1 for a defined time (max. 21 sec) than Clamp_50 = 0

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Test Report

No.	SYSTR5418	Test Case Name:	SYSTS0094
Test type	A	Test Case Outline:	Function test Clamp_50 emergency start
Result:	OK	Tester:	Stefan Pfingstl
OIL References			
Annotations and comments			
Test log			
Starter_Test_Report_1418300024.html			

No.	SYSTR5419	Test Case Name:	SYSTS0794
Test type	M	Test Case Outline:	Reset during engine start
Result:	OK	Tester:	Stefan Pfingstl
OIL References			
Annotations and comments			
Test log			
SYSTR5419.blf			

No.	SYSTR5420	Test Case Name:	SYSTS0795
Test type	A	Test Case Outline:	During engine start defined outputs shall not be activated:
Result:	OK	Tester:	Stefan Pfingstl
OIL References			
Annotations and comments			
Test log			
Starter_Test_Report_1418300024.html			

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Test Log (automated test)

7.12 SYSTS0094: Passed

Test case begin: 2014-08-05 13:16:05 (logging timestamp 7085.140305)
Test case end: 2014-08-05 13:16:19 (logging timestamp 7099.640305)

Preparation of Test Case

Timestamp	Test Step	Description	Result
1. MC_ABAG_1 and other MCs = 15 (due to MC and Checksum Skript): Passed			
7085.140305		Set specified value. CAN signal 'MC_ABAG_1' on bus Chassis-CAN: 15	-
7085.140305		Set specified value. CAN signal 'ALIV_COU_DME' on bus PT-CAN: 15 (Signal ungültig)	-
7085.140305		Set specified value. CAN signal 'ALIV_V' on bus PT-CAN: 15 (Signal invalid)	-
7085.140305		Set specified value. CAN signal 'ALIV_GRB' on bus PT-CAN: 15 (Signal invalid)	-
7085.140305		Set specified value. CAN signal 'ALIV_COU_DSC' on bus PT-CAN: 15 (Signal invalid)	-
7085.140305		Set specified value. CAN signal 'ALIV_TORQ_1_DME' on bus PT-CAN: 15 (Signal invalid)	-
7085.140305		Set specified value. CAN signal 'ALIV_TORQ_3_DME' on bus PT-CAN: 15 (Signal invalid)	-
7085.140305	1	Stimulation of the input parameters	pass
7085.140305		Emergency start	-
2. Start conditions: Passed			
7085.140305		Set specified value. CAN signal 'St_crashOut' on bus Chassis-CAN: 0 (No crash)	-
7085.140305		Set specified value. CAN signal 'ST_ENG_RUN' on bus PT-CAN: 0 (Engine off)	-
7085.140305		Set specified value. CAN signal 'V_VEH' on bus PT-CAN: 20	-
7085.140305		Set specified value. CAN signal 'ST_ILK_STRT' on bus PT-CAN: 0 (Kein Motorstart)	-
7085.140305		Set specified value. CAN signal 'ST_CT_BRPD_DME' on bus PT-CAN: 0 (Brake not pressed)	-
7085.140305		Set specified value. CAN signal 'RPM_ENG' on bus PT-CAN: 0	-

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Test file for test automation

Tests_FRSD1.vxt - Editor

Datei Bearbeiten Format Ansicht ?

```
<?xml version="1.0" encoding="iso-8859-1"?>
<!--Vector Test Automation Editor 2.1.28.0-->
<testmodule title="Tests FRS D1" version="1.0" xmlns="http://www.vector-informatik.de/CANoe/TestModule/1.26">
  <testgroup title="Routing">
    <preparation>
      <set title="Key pos. 1">
        <sysvar name="RelayVBatt" namespace="VTS::M12_Ch16_CL_R">1</sysvar>
        <sysvar name="RelayVBatt" namespace="VTS::M12_Ch14_CL_15_1">0</sysvar>
        <sysvar name="RelayVBatt" namespace="VTS::M12_Ch13_CL_50_IK">0</sysvar>
      </set>
      <wait title="Wait" time="500" />
      <set title="Key pos. 2">
        <sysvar name="RelayVBatt" namespace="VTS::M12_Ch16_CL_R">1</sysvar>
        <sysvar name="RelayVBatt" namespace="VTS::M12_Ch14_CL_15_1">1</sysvar>
        <sysvar name="RelayVBatt" namespace="VTS::M12_Ch13_CL_50_IK">0</sysvar>
      </set>
      <wait title="Wait" time="500" />
    </preparation>
    <testcase title="SYSTS0109" ident="">
      <preparation>
        <vardef name="Cycle" type="int" default="0">1</vardef>
        <set title="MC_IPC_2 = 15">
          <cansignal name="MC_IPC_2">15</cansignal>
        </set>
      </preparation>
      <for_loop title="Cycle" loopvar="Cycle" stopvalue="3" startvalue="0" increment="1">
        <set title="N_TempEx = -40">
          <cansignal name="N_TempEx">-40</cansignal>
        </set>
        <wait title="Wait" time="1s" />
      </for_loop>
      <statechange title="N_TempEx - TEMP_EX = -40" wait="1s">
        <in>
          <cansignal name="N_TempEx">-40</cansignal>
        </in>
        <expected>
          <cansignal name="TEMP_EX">-40</cansignal>
        </expected>
      </statechange>
```

System Testbenches

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Network testbench

- Easy accessible ECUs
- All ECUs are connected with supply and bus communication wires to the test bench
- E/E system tests can be done before vehicle building starts
- Automated tests for diagnostic, bus errors, buslogic, network management, gateway
- Fast hardware replacement
- No influence from wiring harness



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Network testbench – test content

- Flash verification
- check of the raw data
- check of logistic data for/after flash
- check the bus behavior by ECU flash
- switch off bus at ECU flash



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Subsystem (domain) testbench

- Automated validation of the complete function-network
- Automation of the validation steps incl. automated test report generating, by plugging and switching via Robot for all customer-, long-term-, voltage-, diagnosis- and coding tests.
- Automation with 4 engineering tools:
  - TestStand
  - LabView
  - CANoe
  - Katana 4D (Robot)



Breakout Rack

Load Rack
ECU´s, operating elements

Periphery Rack
sensors / actuators

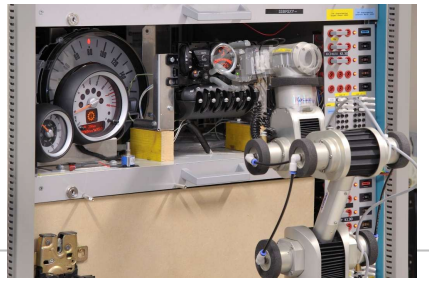
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Subsystem (domain) testbench – test content

System functions (partly automated):

- Pinning / wiring (pinning, coding)
- Bus verification (CAN, LIN, Ethernet, etc.)
  - Bus-Physic (Error Frames, ramp up/down time, voltage level, load resistor, error simulation)
  - Bus-logic (message errors, service messages)
  - Bus load (stress impact, power up/down, rest bus simulation stress load)
- Hardware-Signals 1:n connections
- Voltage tests (under voltage, cranking profile, etc.)
- Failure simulation (signal failure, cable failure, ECU malfunction, invalid signals)
- Misuse tests (reverse polarisation, short cut, etc.)
- Flash, diagnostics
- Gateway tests





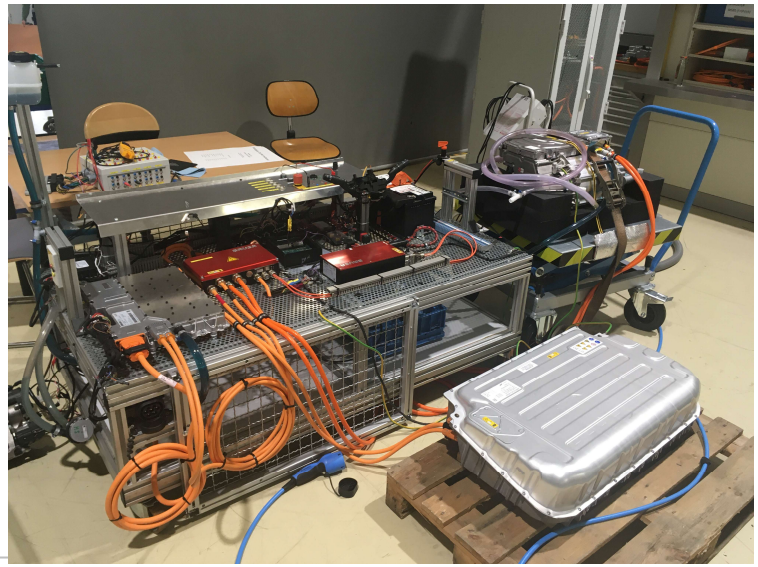
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Powertrain Sub System Test Bench

Test content:

- **Verification of local buses** (CAN-High, CAN-Low, LIN, etc.)
- **Bus-Physic** (Error Frames, ramp up/down time, voltage level, load resistor, error simulation)
- **Energy-Management** (quiescent current, power up/down)
- **Bus-logic** (message errors, service messages)
- **Failure simulation** (signal failure, cable failure, ECU malfunction, invalid signals)
- **Bus load** (stress impact, power up/down, rest bus simulation stress load)
- **Misuse tests**
- **Flash, diagnosis, gateway**
- **Customer function tests** (checklist)
- **HV function validation with all HV components**
- **HW/SW development platform**



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ADAS Sub System Test Bench

Test content:

- Base customer function testing (with original Components)
- Static maneuver testing (Ideal virtual world)
- Failure/Manipulation/Stress testing (Wiring/ECU/Sensor/Actuator failures)
- HMI Validation (functionally)
- EE maturity monitoring
- Vehicle Configuration and Vehicle variant testing
- Safety robustness validation
- Automated testing, Repetitive testing, Long-time testing and Variation testing
- Free and stochastic testing and error analysis

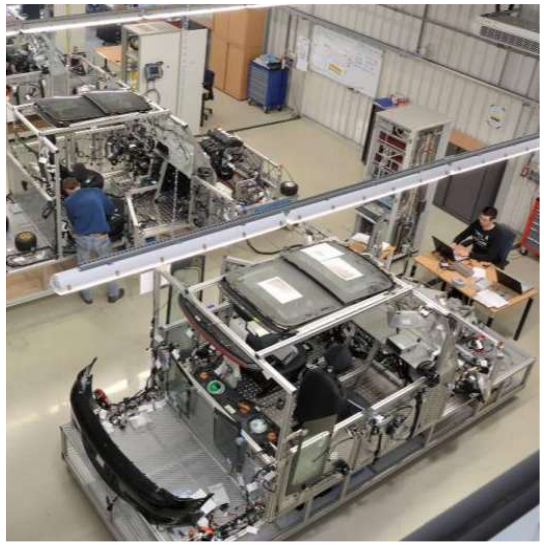


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Laboratory Vehicle (Labcar) - Overview

- Aluminium frame
- Easy accessible ECUs and harness
- Fast hardware replacement without disassembling the complete vehicle
- Original vehicle wiring harness and all ECU´s, sensors and actuators are integrated into the test platform.
- Complete system tests can be done before vehicle building starts
- Automated test sequences (LabVIEW, CANoe, C#)
- HIL-Simulation (NI, dSPACE)
- Residual bus simulation for missing ECU´s





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Laboratory Vehicle (Labcar) – test content

- E/E system function tests
  - wiring harness validation
  - flashing/coding verification
  - physical/logical bus layer tests (CAN,LIN, Ethernet)
  - clamp control tests
  - network management tests (Autosar)
  - hardwired signal measurements
  - voltage tests according specification (ISO 16750-2, LV124,…)
  - quiescent current measurements
  - power management functions
  - diagnostic function tests
  - gateway tests
- Customer function tests





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Vehicle Intensive Test (VIT)

- Assembling and Configuration of special measurement equipment
- Assembling of data logging equipment for AnalysisVehicle upgrades incl. flashing of the control units and error analysis
- Mechanical and electrical Vehicleupgrades
- Commissioning of complete vehicles at the customer and at the PT Shop incl. Flashing, Coding and Functioncheck.







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Vehicle Intensive Test (VIT) – test content

- Validation of system and customer function tests incl. stochastic tests during off-duty testdrives
- “Free testing”
  - Reproduce typical consumer behavior
  - Reproduce misuse
- Summer/wintertestdrives and special component trails on different countries. (e.g. head unit, navigation)
- Support of product demonstrations on trade shows
- Vehicle build up support on the in-plant production line







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Tools: Data Loggers





EMC compliant?	yes	no
Placement close to ECU's?	yes	not always
Interfaces for all bus technologies?	yes	no
Testing without manual?	yes	no
Invisible for the driver?	yes	no
Temperature range (-40° to 60°C)?	yes	no
Shock resistant?	yes	no
Autonomous logging?	yes	no
Second driver needed?	no	sometimes

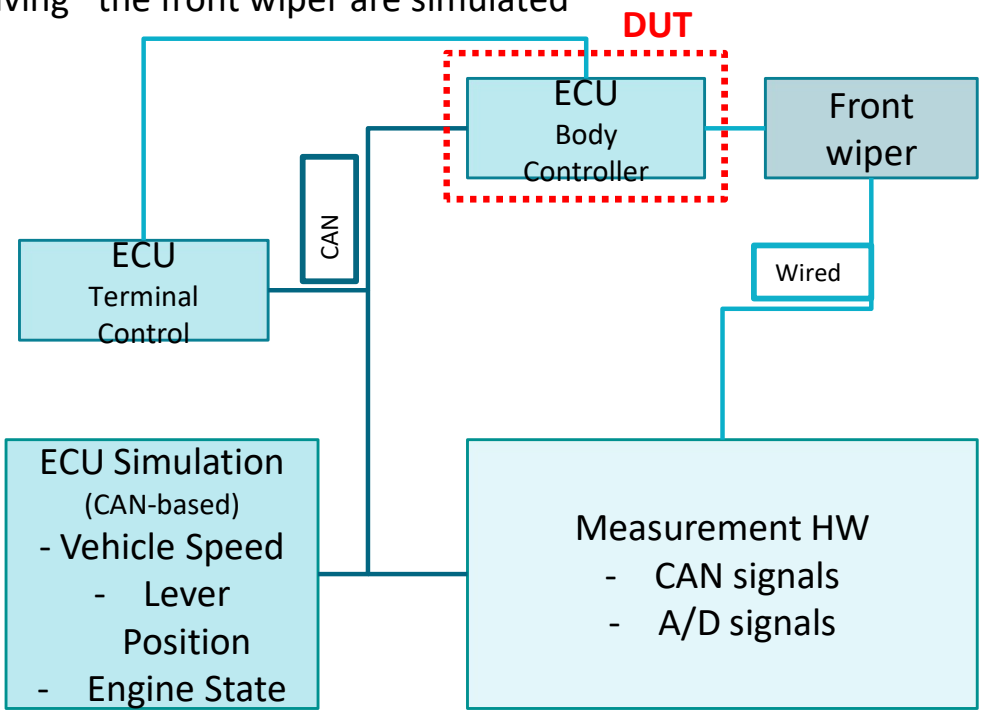
System Testing

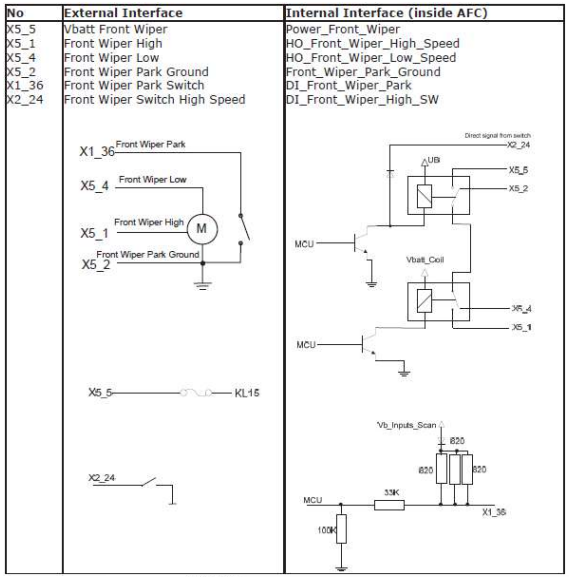
Front Wiper Example

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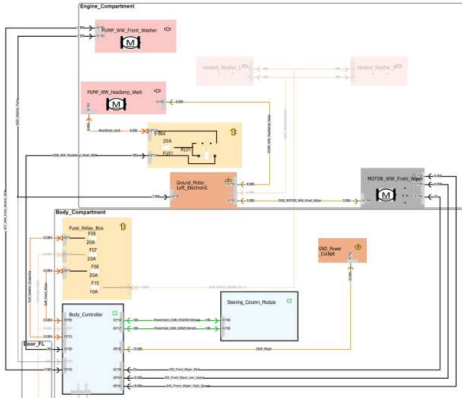
Hardware In the Loop - example

- ECUs which are operating the front wiper assembly are physically available
- ECUs which are not “driving” the front wiper are simulated





No	External Interface	Internal Interface (inside AFC)
X5_5	Vbatt Front Wiper	Power_Front_Wiper
X5_1	Front Wiper High	HO_Front_Wiper_High_Speed
X5_4	Front Wiper Low	HO_Front_Wiper_Low_Speed
X5_2	Front Wiper Park Ground	Front_Wiper_Park_Ground
X1_36	Front Wiper Park Switch	DI_Front_Wiper_Park
X2_24	Front Wiper Switch High Speed	DI_Front_Wiper_High_SW



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Front wiper testing – ECU Simulation

- Engine management system (EMS) simulation for Engine State
- Electronic stability control (ESC) simulation for Vehicle Speed
- Steering column modules (SCM) simulation for Lever Position

From specification

5.3.2 Interface Requirements

5.3.2.2 CAN Signal Inputs

Signal Logical Name	Description
C_FrontWiper_LowSpeed_SW_ST	Front wiper low speed switch input on CAN **- SCM**
C_FrontWiper_HighSpeed_SW_ST	Front wiper high speed switch input on CAN **- SCM**
C_FrontWiper_IntermSpeed_SW_ST	Front wiper intermittent speed switch input on CAN **- SCM**
C_FrontWiper_MistMode_SW_ST	Front wiper mist mode switch input on CAN - **SCM**
C_FrontWiper_Interval_SW_ST	Front wiper interval switch input on CAN **- SCM**
C_FrontWasher_SW_ST	Front washer switch input on CAN **- SCM**
C_VehicleSpeed_ESC	Speed information from ESC ECU**- ESC**
C_EngineState_EMS_ST	Engine state status from EMS ECU **- EMS**

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Front wiper testing – Measurement

- A/D signals to monitor
- CAN signals

From specification

5.3.2.2 Digital Inputs

Signal Logical Name	Description
HW_Ignition_ST	KL15 power supply input
HW_FrontWiper_Park_SW_ST	Front wiper park switch status

5.3.2.3 Digital Outputs

Signal Logical Name	Description
HW_FrontWiper_OnOff_CMD	Front wiper power enable relay output
HW_FrontWiper_HighLow_CMD	Front wiper high/low speed relay output
HW_FrontWasher_CMD	Front washer output
HW_HeadlampWasher_CMD	Headlamp washer output

Front wiper testing – Speed Mode Test

From specification

5.3.5 Function Requirements

5.3.5.4 Front Wiper Intermittent Mode

Dwell Time of Front Wiper Intermittent Speed Class Mode
ID: FRS_AFC_FWP_018:
There are 5 speed class variations available with 4 dwell times in each class. The dwell time should only be changed from one vehicle speed class to another vehicle speed class when the wiper motor is in park position.

Interval Adjust switch position	Vehicle speed class (kph)				
	V0 (<5)	V1 (32)	V2 (<64)	V3 (<96)	V4 (>=96)
Interval 1, dwell time in s	24	16	12	10	8
Interval 2, dwell time in s	15	10	8	6	4
Interval 3, dwell time in s	8	5	4	3	2
Interval 4, dwell time in s	3	2	2	1	1
Default, dwell time in s	8	5	4	3	2

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Wiper Speed Mode Testing – Test sequence

- Prepare test environment (supply voltage, parameter sets, simulated ECUs…)
- Prepare test modules
- Start testing
- Test report generation

```
variables
{
// ms Timer
  msTimer mst_EMS_1;                    // 10ms
  msTimer mst_EMS_2;                    // 10ms
  msTimer mst_EMS_3;                    // 10ms
  msTimer mst_EMS_4;                    // 50ms
  msTimer mst_EMS_5;                    // 100ms
  msTimer mst_EMS_7;                    // 10ms

// Messages
  Message FRSD0_Powertrain_CAN::EMS_1 msEMS_1;
  Message FRSD0_Powertrain_CAN::EMS_2 msEMS_2;          // ST
  Message FRSD0_Powertrain_CAN::EMS_3 msEMS_3;          // En
  Message FRSD0_Powertrain_CAN::EMS_4 msEMS_4;          // ST
  Message FRSD0_Powertrain_CAN::EMS_5 msEMS_5;          // Le
  Message FRSD0_Powertrain_CAN::EMS_6 msEMS_6;          // no
  Message FRSD0_Powertrain_CAN::EMS_7 msEMS_7;


    int engine_state, stop_condition, crank_condition, engin
```

Test Overview

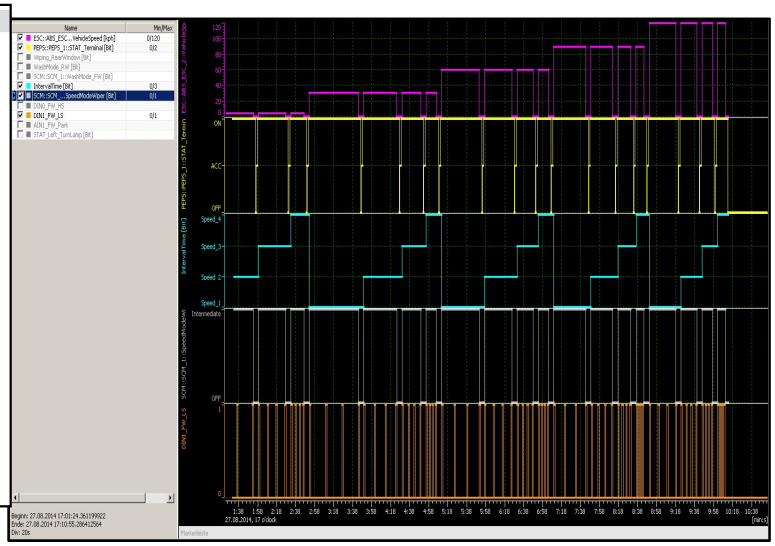
Test begin: 2014-08-27 16:59:40 (logging timestamp 1.5070)
Test end: 2014-08-27 17:10:14 (logging timestamp 635.3947)

Statistics

Overall number of test cases	20	
Executed test cases	20	100% of all test cases
Not executed test cases	0	0% of all test cases
Test cases passed	20	100% of executed test cases
Test cases failed	0	0% of executed test cases

Test Case Results

1	FRS_AFC_FWP_018	Lever in intermittent mode position with interval mode 1 and speed class 0	pass
2	FRS_AFC_FWP_018	Lever in intermittent mode position with interval mode 2 and speed class 0	pass
3	FRS_AFC_FWP_018	Lever in intermittent mode position with interval mode 3 and speed class 0	pass
4	FRS_AFC_FWP_018	Lever in intermittent mode position with interval mode 4 and speed class 0	pass
5	FRS_AFC_FWP_018	Lever in intermittent mode position with interval mode 1 and speed class 1	pass
6	FRS_AFC_FWP_018	Lever in intermittent mode position with interval mode 2 and speed class 1	pass
7	FRS_AFC_FWP_018	Lever in intermittent mode position with interval mode 3 and speed class 1	pass
8	FRS_AFC_FWP_018	Lever in intermittent mode position with interval mode 4 and speed class 1	pass
9	FRS_AFC_FWP_018	Lever in intermittent mode position with interval mode 1 and speed class 2	pass
10	FRS_AFC_FWP_018	Lever in intermittent mode position with interval mode 2 and speed class 2	pass
11	FRS_AFC_FWP_018	Lever in intermittent mode position with interval mode 3 and speed class 2	pass
12	FRS_AFC_FWP_018	Lever in intermittent mode position with interval mode 4 and speed class 2	pass
13	FRS_AFC_FWP_018	Lever in intermittent mode position with interval mode 1 and speed class 3	pass
14	FRS_AFC_FWP_018	Lever in intermittent mode position with interval mode 2 and speed class 3	pass
15	FRS_AFC_FWP_018	Lever in intermittent mode position with interval mode 3 and speed class 3	pass
16	FRS_AFC_FWP_018	Lever in intermittent mode position with interval mode 4 and speed class 3	pass
17	FRS_AFC_FWP_018	Lever in intermittent mode position with interval mode 1 and speed class 4	pass



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Hardware In the Loop - Tools used

- **vTESTstudio** - development environment for creating automated ECU tests
- **CANoe** - software tool for development, test, and analysis of entire ECU networks and individual ECUs (LINK)
- **VT System** - Modular Test Hardware comprising bus interfaces, measurement modules, stimulation modules, relay modules…

